

Do Tendons Matter? Controllability of a Soft-PVTOL with Passive Elastic Arms

Gerardo Flores, Rodolfo Verdín, and Mark W. Spong

Abstract—We ask whether tendon actuation is necessary to fly a soft-arm aerial vehicle. In the Soft-PVTOL, a planar platform whose rigid arms are replaced by soft continuum members, tendons nominally control arm curvature, giving four inputs and five degrees of freedom. With the tendons inactive, the system retains only two thrust inputs while gaining unactuated elastic dynamics, raising a controllability question with no counterpart in rigid multirotor theory. For the physical parameter set of the prototype, we establish local controllability at hover at every admissible equilibrium outside an isolated exceptional stiffness, at which the linearization remains stabilizable, with persistence under sufficiently small parameter perturbations. The proof exploits the system’s bilateral symmetry to split the linearized dynamics into two SISO subsystems, each verified by the Popov–Belevitch–Hautus test; the enabling mechanism is the thrust-induced generalized force $T_{il_i}(1 - \cos q_i)/q_i^2$, the sole actuation path to the arm curvatures. Simulations under asymmetric arm parameters confirm stable hover from thrust alone, while test-stand experiments with disabled tendons validate accurate pitch tracking and passive curvature convergence. Tendon-free operation is therefore a viable fallback, at the cost of reduced control authority.

Index Terms—Soft aerial vehicle, Soft PVTOL, Soft robotics, controllability.

I. INTRODUCTION

Morphing aerial platforms seek to increase the versatility of multirotors by adapting their geometry to the task or environment. Existing designs include foldable [1], origami-inspired [2], [3], tilt-rotor [4], and reconfigurable-arm [5], [6] vehicles. Most of these platforms, however, achieve reconfiguration through rigid joints and mechanical linkages, rather than through structurally compliant members [7]–[9].

Soft and continuously deformable arms offer an alternative way to embed morphing directly into the vehicle structure. In this direction, Flores and Spong introduced the *Soft-PVTOL* [10], [11], where the rigid arms of the classical PVTOL [12] are replaced by soft continuum arms modeled under

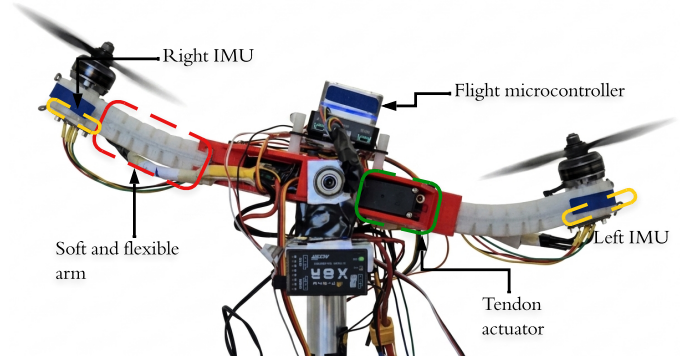


Fig. 1: Experimental Soft-PVTOL prototype on the test stand, showing the thrust-induced deflection of the flexible arms in passive mode (tendon actuators disabled). IMUs at each arm tip provide state estimation.

the constant-curvature assumption [13]. In the nominal design, tendons actively control each arm’s curvature, giving four inputs (two thrusts and two tendon torques) for five degrees of freedom. The planar Soft-PVTOL prototype of Fig. 1 is used for the experimental validation reported here; Section V-B discusses what the planar reduction leaves out, and free-flight recordings of the QuadSoft, its three-dimensional counterpart, are supplementary, qualitative support only.

Related soft and compliant aerial vehicles have primarily used passive arm deformation for collision tolerance, impact resilience, and morphing [14]–[17]. In contrast, we ask whether the Soft-PVTOL can remain controllable when its tendon actuators are disabled: the arms then evolve as passive elastic elements, coupled to the body only through propeller thrust, and the system changes from a 5-DOF, four-input vehicle to a 5-DOF, two-input one. We show that tendon-free operation preserves controllability, albeit with reduced maneuverability, making it a viable fallback rather than the nominal mode.

Prior work on actuator loss in multirotors assumes a *rigid airframe*: the geometry is invariant under failure, and controllability reduces to the rank of a fixed allocation matrix [18]–[21]. Here the system does not simply lose inputs; it simultaneously gains unactuated, nonlinear arm dynamics coupled to the body only through the thrusts. Whether the two remaining thrust inputs can stabilize both the rigid-body motion *and* the arm deformations is a nontrivial question with no counterpart in the rigid multirotor literature.

To our knowledge, this is the first work that addresses the controllability of a soft aerial vehicle with passive, tendon-free arms. The contributions of this paper are:

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This letter has supplementary downloadable material: videos of the passive-arm test-stand validation of Section V and of QuadSoft free flight with deactivated tendons, including an in-flight deactivation test. Available at <https://github.com/gfloresc/soft-pvtol-passive-arms>.

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- 1) We derive the passive-arm model and show that the thrust-induced generalized force $T_i l_i (1 - \cos q_i)/q_i^2$ provides the sole actuation path from the propeller thrusts to the passive arm-curvature dynamics.
- 2) For the considered passive-arm Soft-PVTOL, we establish local controllability of the system linearization about every admissible hover equilibrium outside an isolated exceptional stiffness, at which the linearization remains stabilizable, with robustness to sufficiently small parameter perturbations. The proof combines a symmetry-based decomposition into two SISO subsystems with the PBH test.
- 3) We validate the results through simulations, a stiffness sweep of the PBH rank condition, and test-stand experiments.

II. THE SOFT-PVTOL MODEL

A. System Description

The Soft-PVTOL of [10], whose passive-arm variant we derive below, consists of a rigid body of mass m and moment of inertia I , connected to two soft continuum arms of lengths l_l and l_r carrying a tip mass m_i and a lumped rotational inertia I_i conjugate to the bending rate $\dot{q}_i$; both enter $D(\mathbf{q})$. Under the constant curvature assumption [13], each arm deforms into a circular arc parameterized by its total bending angle q_i , where the index $i \in \{l, r\}$ denotes the left and right arm, respectively, so that q_l and q_r are the two arm-curvature coordinates. The generalized coordinates are¹:

$$\mathbf{q} = (x_v, z_v, \theta, q_l, q_r)^\top \in \mathbb{R}^5, \quad \mathbf{u} = (T_l, T_r)^\top \in \mathbb{R}^2. \quad (1)$$

The dynamics follow the standard form [10]:

$$D(\mathbf{q})\ddot{\mathbf{q}} + C(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}} + g(\mathbf{q}) = \boldsymbol{\tau}, \quad (2)$$

where $D(\mathbf{q}) \in \mathbb{R}^{5 \times 5}$ is the symmetric, positive-definite inertia matrix, $C(\mathbf{q}, \dot{\mathbf{q}})$ is the Coriolis matrix ($\dot{D} - 2C$ is skew-symmetric), and $g(\mathbf{q})$ is the gravity vector. These matrices involve the parameter groupings $\Theta_1 = m + m_l + m_r$, $\Theta_3 = l_l m_l$, $\Theta_4 = l_r m_r$, and contain expressions such as $\sin(q_i)/q_i$ that require evaluation via their Taylor series limits as $q_i \rightarrow 0$: $\text{sinc}(q) \triangleq \frac{\sin q}{q} \rightarrow 1$, $\frac{1 - \cos q}{q} \rightarrow 0$, $\frac{1 - \cos q}{q^2} \rightarrow \frac{1}{2}$. The complete expressions for D , C , and g are as in [10].

B. Generalized Forces and the Tendon-Free System

The thrust forces produce the first three components of the generalized force vector:

$$\begin{aligned} \tau_x &= \cos \theta (-T_r \sin q_r + T_l \sin q_l) - \sin \theta (T_r \cos q_r + T_l \cos q_l) \\ \tau_z &= \sin \theta (-T_r \sin q_r + T_l \sin q_l) + \cos \theta (T_r \cos q_r + T_l \cos q_l) \\ \tau_\theta &= l_r T_r \text{sinc}(q_r) - l_l T_l \text{sinc}(q_l), \end{aligned} \quad (3)$$

where $\tau_\theta = \sum_i p_i \times f_i$ is the moment of the tip thrusts about the body center, with $p_i = l_i [\text{sinc } q_i, (1 - \cos q_i)/q_i]^\top$ and f_i normal to the tip tangent. These generalized forces are common to both the tendon-actuated and the tendon-free configurations. In the original tendon-actuated Soft-PVTOL [10],

two additional generalized forces, the tendon torques τ_l^t and τ_r^t , act as independent control inputs that directly command the arm curvatures, for four inputs in total. In this work we consider *the tendon-free system*, in which these tendon torques are absent and the arm curvatures are left passive. In this case the arm curvatures evolve passively under an elastic restoring torque $-k_i q_i$, viscous damping $-b_i \dot{q}_i$, and the thrust. Because q_i parameterizes the curvature of the arc rather than a hinge angle, the generalized force of the tip thrust is the projection $Q_{q_i} = f_i^\top \partial p_i / \partial q_i = T_i l_i (1 - \cos q_i)/q_i^2$, and gravity enters $g(\mathbf{q})$ by the same route, through $\partial p_{i,z} / \partial q_i = l_i G_4(q_i)$ with G_4 defined in Section III-A.

The generalized force vector becomes $\boldsymbol{\tau}_{\text{passive}} = [\tau_x, \tau_z, \tau_\theta, \tau_{q_l}, \tau_{q_r}]^\top$, with $\tau_x, \tau_z, \tau_\theta$ as in (3) and the passive arm torques

$$\tau_{q_i} = -k_i q_i - b_i \dot{q}_i + T_i l_i \frac{1 - \cos q_i}{q_i^2}, \quad i \in \{l, r\}. \quad (4)$$

The system now has only two control inputs, $\mathbf{u} = [T_l, T_r]^\top$. The last term in each arm equation, $T_i l_i (1 - \cos q_i)/q_i^2$, is the *only* mechanism through which they influence the arm curvatures in the absence of tendons; the arm coordinates are otherwise unactuated.

This raises the central question: *is the Soft-PVTOL with passive elastic arms controllable using only the thrust inputs?*

III. CONTROLLABILITY ANALYSIS

We analyze the system about hover, the nominal operating point of multirotor vehicles. For analytical clarity we first take identical arm parameters ($l_l = l_r = l$, $k_l = k_r$, $m_l = m_r$), which enables a decomposition into symmetric and antisymmetric modes; Remark 1 treats the mismatched case.

A. Hover Equilibrium

At hover ($\dot{\mathbf{q}} = \ddot{\mathbf{q}} = 0$, $\theta = 0$), the equilibrium condition $g(\mathbf{q}) = \boldsymbol{\tau}$ yields, for the symmetric case ($l_l = l_r = l$, $k_l = k_r = k$, $m_l = m_r$):

$$T^* = \frac{g\Theta_1}{2 \cos q^*}, \quad (5)$$

$$k q^* = T^* \cdot l \cdot \frac{1 - \cos q^*}{(q^*)^2} - g \Theta_3 \cdot G_4(q^*), \quad (6)$$

where $G_4(q) = (q \sin q + \cos q - 1)/q^2$. Several observations follow. Whenever an admissible equilibrium exists, its curvature satisfies $q^* > 0$, so the arms are never straight at hover: the thrust-induced torque deflects them into a convex shape as seen from the front, with the rotor-carrying tips raised relative to the body anchor, as visible in the passive-arm configurations of Figs. 1 and 3, and this deflection is balanced by the elastic restoring force. Consequently, from (5), the hover thrust satisfies $T^* > g\Theta_1/2$ since $\cos q^* < 1$: the deflected arms waste a portion of the thrust laterally and require more total thrust than the rigid PVTOL. Finally, as $k \rightarrow \infty$ the arms stiffen, $q^* \rightarrow 0$ and $T^* \rightarrow g\Theta_1/2$, recovering the classical PVTOL hover.

¹A boldface $\mathbf{q}$ denotes the full state vector; an italic q without subscript is a generic bending angle, and scalar functions such as $\text{sinc}(q)$ act componentwise on q_l, q_r .

Linearizing the Euler–Lagrange dynamics (2) around the symmetric hover equilibrium

$$\mathbf{q}^* = (0, z_0, 0, q^*, q^*)^\top, \quad \mathbf{u}^* = (T^*, T^*)^\top, \quad (7)$$

where $\mathbf{q}^* \in \mathbb{R}^5$ and $\mathbf{u}^* \in \mathbb{R}^2$ are the equilibrium state and input vectors, while the scalars $q^* > 0$ and $T^* > 0$ are, respectively, the common equilibrium arm curvature and the common per-rotor equilibrium thrust that satisfy the hover conditions (5)–(6) (at the symmetric equilibrium both arms share the same curvature q^* and both rotors the same thrust T^*), yields the second-order LTI model

$$M^* \ddot{\delta \mathbf{q}} + D^* \dot{\delta \mathbf{q}} + K^* \delta \mathbf{q} = B^* \delta \mathbf{u}, \quad (8)$$

with $M^* \succ 0$, and where K^* denotes the linearization of the generalized position-dependent forces (gravity + elasticity + thrust-induced terms) at hover. D^* contains only the arm damping, since the Coriolis term is quadratic in the velocities; and K^* is in general non-symmetric, since the thrust is a follower force and its contribution does not derive from a potential.

B. Bilateral symmetry and exact decoupling

The controllability analysis exploits the bilateral symmetry of the Soft-PVTOL when the arms have identical mechanical properties. Let us define the bilateral symmetry (reflection about the vertical axis) as: $(x_v, z_v, \theta, q_l, q_r) \mapsto (-x_v, z_v, -\theta, q_r, q_l)$, and let $S \in \mathbb{R}^{5 \times 5}$ be its induced linear action at hover. Since K^* is not derived from a potential, invariance of the Lagrangian alone does not suffice. What holds is that the *forced* equations are equivariant under the reflection together with the interchange $T_l \leftrightarrow T_r$; linearizing at the symmetric equilibrium, where $q_l^* = q_r^*$ and $T_l^* = T_r^*$, then gives

$$SM^* = M^*S, \quad SD^* = D^*S, \quad SK^* = K^*S, \quad (9)$$

and the input matrix transforms by swapping left/right thrust channels:

$$SB^* = B^*P, \quad P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}. \quad (10)$$

Let us introduce symmetric/antisymmetric coordinates and inputs

$$\begin{aligned} q_s &= \frac{q_l + q_r}{2}, & q_a &= \frac{q_r - q_l}{2}, \\ T_s &= T_l + T_r, & T_d &= T_r - T_l, \end{aligned} \quad (11)$$

and the transformed vectors

$$\delta \tilde{\mathbf{q}} = (x_v, z_v, \theta, q_s, q_a)^\top, \quad \delta \tilde{\mathbf{u}} = (\delta T_s, \delta T_d)^\top. \quad (12)$$

With $\delta \mathbf{q} = T_q \delta \tilde{\mathbf{q}}$ and $\delta \mathbf{u} = T_u \delta \tilde{\mathbf{u}}$, where T_q acts as the identity on (x_v, z_v, θ) and as $\begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$ on (q_s, q_a) , and $T_u = \frac{1}{2} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$, the transformed model follows from the congruence that preserves virtual work, $\tilde{M} = T_q^\top M^* T_q$, $\tilde{D} = T_q^\top D^* T_q$, $\tilde{K} = T_q^\top K^* T_q$, $\tilde{B} = T_q^\top B^* T_u$. Using (9)–(10) it is block diagonal:

$$\tilde{M} \ddot{\delta \tilde{\mathbf{q}}} + \tilde{D} \dot{\delta \tilde{\mathbf{q}}} + \tilde{K} \delta \tilde{\mathbf{q}} = \tilde{B} \delta \tilde{\mathbf{u}}, \quad (13)$$

where $\tilde{M} = \begin{bmatrix} M_a & 0 \\ 0 & M_s \end{bmatrix}$, $\tilde{B} = \begin{bmatrix} 0 & b_a \\ b_s & 0 \end{bmatrix}$, and the antisymmetric block corresponds to (x_v, θ, q_a) driven only by δT_d , and the

symmetric block corresponds to (z_v, q_s) driven only by δT_s . Equivalently, the system decomposes into two independent SISO subsystems:

$$\Sigma_s : M_s \ddot{\xi}_s + D_s \dot{\xi}_s + K_s \xi_s = b_s \delta T_s, \quad (14)$$

$$\Sigma_a : M_a \ddot{\xi}_a + D_a \dot{\xi}_a + K_a \xi_a = b_a \delta T_d, \quad (15)$$

where $\xi_s = (z_v, q_s)^\top \in \mathbb{R}^2$, and $\xi_a = (x_v, \theta, q_a)^\top \in \mathbb{R}^3$.

Lemma 1 (Block-diagonal controllability). *Let (A, B) be the first-order realization of (8). Under the transformation (11), (A, B) is similar to a block-diagonal pair $(\tilde{A}, \tilde{B})$ whose blocks correspond to (14) and (15). Hence, the full linearization is controllable if and only if both subsystems Σ_s and Σ_a are controllable.*

Proof. The transformation $\delta \mathbf{q} = T_q \delta \tilde{\mathbf{q}}$, $\delta \mathbf{u} = T_u \delta \tilde{\mathbf{u}}$ induces, through the congruence above, a similarity transformation on the first-order realization of (8) composed with an invertible input transformation; neither alters controllability. Because the matrices commute with S as in (9) and the input is equivariant as in (10), the transformed matrices simultaneously preserve the symmetric/antisymmetric eigenspaces of S , yielding the block form (13); therefore the full pair is controllable iff each independent block is controllable. $\square$

C. Controllability of the SISO subsystems

We use the Popov–Belevitch–Hautus (PBH) test written for second-order systems. Consider

$$M \ddot{\xi} + D \dot{\xi} + K \xi = b u, \quad M \succ 0. \quad (16)$$

Let $P(\lambda) = \lambda^2 M + \lambda D + K$ be the quadratic pencil. The first-order realization of (16) is uncontrollable if and only if there exist $\lambda \in \mathbb{C}$ and a left null vector $w \neq 0$ such that $w^\top P(\lambda) = 0$, $w^\top b = 0$.

a) *Input directions:* At hover with $q^* \in (0, \pi/2)$, the collective and differential thrust directions are $b_s = [\cos q^*, l(1 - \cos q^*)/q^{*2}]^\top$ and $b_a = [-\sin q^*, l \operatorname{sinc}(q^*), l(1 - \cos q^*)/q^{*2}]^\top$, whose entries are all strictly positive in magnitude whenever $q^* \in (0, \pi/2)$ (Lemma 2).

Lemma 2 (Admissible hover equilibrium). *Consider the symmetric hover equilibrium (5)–(6) and define the scalar equilibrium map $k = \kappa(q)$ obtained by substituting $T^*(q) = g\Theta_1/(2 \cos q)$ from (5) into (6),*

$$\kappa(q) = \frac{g}{q^3} \left[\frac{\Theta_1 l (1 - \cos q)}{2 \cos q} - \Theta_3 (q \sin q + \cos q - 1) \right], \quad (17)$$

where $q \in (0, \frac{\pi}{2})$. Then κ is continuous and positive on $(0, \pi/2)$ and satisfies $\kappa(q) \rightarrow +\infty$ as $q \rightarrow 0^+$ and as $q \rightarrow (\pi/2)^-$. Consequently κ attains a positive interior minimum $k_{\min} := \min_{q \in (0, \pi/2)} \kappa(q)$ at some $q_m \in (0, \pi/2)$, and for every $k > k_{\min}$ the equation $\kappa(q) = k$ has at least one solution on the low-curvature branch $(0, q_m)$. Any such solution satisfies $q^* \in (0, \pi/2)$, and therefore $\cos q^* > 0$, $\sin q^* > 0$, $\operatorname{sinc}(q^*) > 0$, and $(1 - \cos q^*)/q^{*2} > 0$.

Proof. Continuity on $(0, \pi/2)$ is immediate from (17). For positivity, note that $h(q) = (1 - \cos q) - (q \sin q + \cos q - 1)$ satisfies $h(0) = 0$ and $h'(q) = \sin q - q \cos q \geq 0$ on $(0, \pi/2)$,

so $q \sin q + \cos q - 1 \leq 1 - \cos q$; together with $1/\cos q \geq 1$ the bracket is bounded below by $(1 - \cos q)(\frac{1}{2}\Theta_1 l - \Theta_3) > 0$, because the total mass times arm length exceeds twice the arm first moment ($\Theta_1 l - 2\Theta_3 = l(m + 2m_l) - 2lm_l = lm > 0$, i.e. simply that the rigid body has positive mass). For the boundary behaviour, as $q \rightarrow 0^+$ the bracket is $\frac{1}{4}(\Theta_1 l - 2\Theta_3) q^2 + O(q^4)$ so $\kappa(q) = \frac{g(\Theta_1 l - 2\Theta_3)}{4q} + O(q) \rightarrow +\infty$; as $q \rightarrow (\pi/2)^-$, $1/\cos q \rightarrow +\infty$ gives $\kappa(q) \rightarrow +\infty$. A positive continuous function diverging at both ends of $(0, \pi/2)$ attains an interior minimum $k_{\min} = \kappa(q_m) > 0$, and by the intermediate value theorem $\kappa(q) = k$ is solvable on $(0, q_m)$ for every $k > k_{\min}$. Any such solution has $q^* \in (0, \pi/2)$, giving the stated strict positivities. Uniqueness and strict monotonicity of κ on $(0, q_m)$ hold in the physical parameter range and are verified numerically. $\square$

Theorem 1 (Controllability at admissible hover equilibria). *Consider the linearization (8) at an admissible hover equilibrium $q^* \in (0, \pi/2)$, which by Lemma 2 exists for every $k > k_{\min}$, with the mass, length and inertia values of Table I. Let $\Delta_\bullet(k, b)$ denote the Kalman determinant of $\Sigma_\bullet$, $\bullet \in \{s, a\}$. Then:*

- (i) Δ_s has no zero on $q^* \in (0, \pi/2)$, $b \in [0.05, 3]$ N m s/rad, so Σ_s is controllable throughout that range;
- (ii) Σ_a is controllable at every admissible (k, b) except on the one-dimensional set $\mathcal{E} = \{\Delta_a = 0\}$, on which the arm-relaxation pole coincides with a transmission zero and becomes unreachable from δT_d . For the prototype damping $b = 0.5$ N m s/rad, $\mathcal{E}$ is crossed at $k_{\text{exc}} \approx 1.17$ N m/rad ($q^* \approx 11.5^\circ$), while the prototype operates at $k = 2.45$;
- (iii) on $\mathcal{E}$ the unreachable mode is stable, so the linearization is stabilizable at every admissible equilibrium examined; selecting k away from $\mathcal{E}$ is therefore a design guideline, not a feasibility constraint, and the prototype does so with a factor of two.

Hence the full linearization is controllable for the prototype and for all admissible $(k, b) \notin \mathcal{E}$, and by openness this persists on a neighborhood of any such point.

Proof. By Lemma 1, controllability of the linearization is equivalent to controllability of Σ_s and Σ_a . For a single-input second-order pair (M, D, K, b) , the PBH criterion states that the pair is controllable iff $\text{rank}[\lambda^2 M + \lambda D + K \mid b]$ is full for every $\lambda \in \mathbb{C}$; a rank drop at some λ requires a left null vector $w \neq 0$ with $w^\top(\lambda^2 M + \lambda D + K) = 0$ and $w^\top b = 0$, equivalently $\Delta_\bullet = 0$. All entries of $M_\bullet, D_\bullet, K_\bullet, b_\bullet$ are real-analytic in (q^*, b) on the connected domain $(0, \pi/2) \times (0, \infty)$, hence so is $\Delta_\bullet$; since $\Delta_s, \Delta_a \neq 0$ at the prototype point, neither vanishes identically, their zero sets have measure zero, and for fixed b the zeros in q^* are isolated. By Lemma 2, the same holds in the (k, b) plane. By Lemma 2, b_s and b_a have all entries nonzero at every admissible q^* , the arm entry $l(1 - \cos q^*)/q^{*2}$ being the thrust-induced generalized force of (4); this is necessary for, but does not by itself imply, full rank, which is verified on the pencils.

(i) Δ_s has no sign change on the stated range and the PBH matrix of Σ_s stays full rank at every eigenvalue of its pencil.

(ii) Δ_a changes sign exactly once on the low-curvature branch for each b , at $k_{\text{exc}} \approx 0.44, 0.84, 1.17, 1.85, 2.93$ N m/rad for $b = 0.1, 0.3, 0.5, 1.0, 2.0$ N m s/rad; there the pencil is rank-deficient at a single real eigenvalue, the arm-relaxation pole, which approaches $-k/b$ in the overdamped limit (within 5% over $b \in [0.1, 2]$).

(iii) At every point examined, including those on $\mathcal{E}$, the pencil has full rank at $\lambda = 0$ and at the unstable real eigenvalue of Σ_a , so only a stable mode is lost on $\mathcal{E}$.

Openness of controllability yields persistence on a neighborhood of any admissible $(k, b) \notin \mathcal{E}$, in particular under sufficiently small left–right parameter mismatches. $\square$

Remark 1 (Robustness to arm-parameter mismatch). *The exact symmetric/antisymmetric decoupling relies on identical arm parameters ($k_l = k_r, m_l = m_r, l_l = l_r$); small mismatches break the invariance and couple the two modal blocks. By the openness argument in the proof of Theorem 1, the controllability result nevertheless persists under sufficiently small left–right variations.*

D. Dependence on the stiffness parameter

The arm stiffness k enters through the elastic potential energy, modifying both $K^*(k)$ and the equilibrium $q^*(k)$. By Lemma 2 and Theorem 1, controllability therefore holds across the whole admissible range $k > k_{\min}$ up to the exceptional set $\mathcal{E}$, not only near a single value. For Table I, κ is monotone decreasing up to $q_m \approx 50^\circ$, where $k_{\min} \approx 0.41$ N m/rad — well below the stiffness of typical soft arms, so the prototype and every configuration considered here lie comfortably inside the admissible range.

IV. CONTROL AND SIMULATION

A. Control Strategy

The simulations provide a constructive check of Theorem 1 rather than an optimized control design. With only two inputs (T_l, T_r) , we use the classical cascaded PVTOL structure, in which horizontal motion is produced indirectly through pitch. Altitude uses a PID law with feedforward gravity compensation and anti-windup, $F_z = g\Theta_1 + k_{p,z}(z_d - z_v) + k_{d,z}(\dot{z}_d - \dot{z}_v) + k_{i,z} \text{sat}(i_z)$, with a smooth C^1 quintic takeoff reference held to avoid an unrealistic instantaneous climb-rate command at $z_d = 5$ m. Horizontal position uses a PID law, whose integral term removes the small steady-state offset caused by the lateral bias of the asymmetric passive arms, $F_x = k_{p,x}(x_d - x_v) + k_{d,x}(-\dot{x}_v) + k_{i,x} \int (x_d - x_v) dt$, from which the desired pitch is $\theta_d = -\arctan(F_x / \max(F_z, F_{\min}))$, clipped to $|\theta_d| \leq 0.4$ rad. Pitch is regulated by a PD law $\tau_{\theta,d} = k_{p,\theta}(\theta_d - \theta) - k_{d,\theta}\dot{\theta}$, and the thrusts are allocated as $T_r = (Fl_l + \tau_{\theta,d})/(l_l + l_r)$, $T_l = F - T_r$ with $F = \sqrt{F_x^2 + F_z^2}$, both clipped to $[0, T_{\max}]$, $T_{\max} = 1.8 T_{\text{hover}}$ with $T_{\text{hover}} = g\Theta_1/2$ (Table I). The arm curvatures (q_l, q_r) are not controlled; they evolve as unactuated dynamics governed by (4). Gains (k_p, k_d, k_i) : $(8, 4, 1.5)$, $(1.5, 2, 0.4)$, $(6, 2)$ for altitude, position, and attitude.

TABLE I: Physical parameters of the Soft-PVTOL, derived from the experimental prototype. Stiffness is estimated from the TPU flexural modulus and arm cross-section; damping is identified from flight data.

Parameter	Value	Parameter	Value
Body mass m	0.853 kg	Arm length l	0.110 m
Arm mass m_l, m_r	0.064 kg	Body inertia I	$3.4 \times 10^{-3} \text{ kg m}^2$
Arm inertia I_l, I_r	$5.1 \times 10^{-4} \text{ kg m}^2$	Stiffness k	2.45 N m/rad
Damping b	0.5 N m s/rad	Hover T_{hover}	4.80 N
Sat. T_{max}	8.64 N	Total mass	0.98 kg

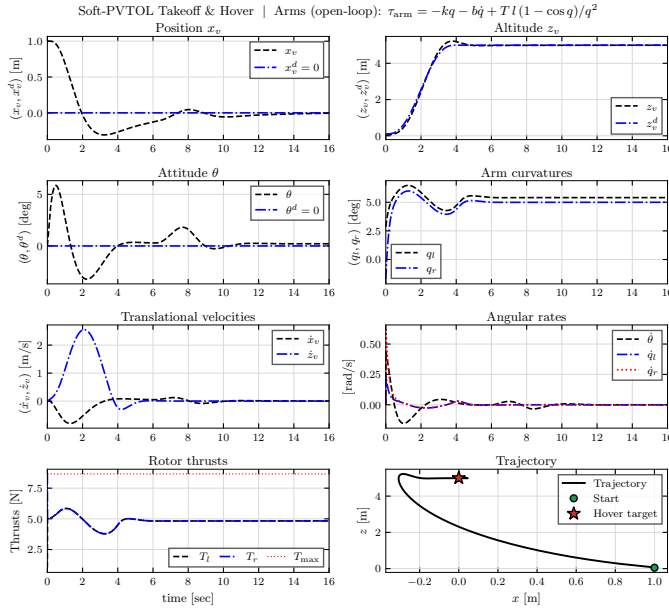


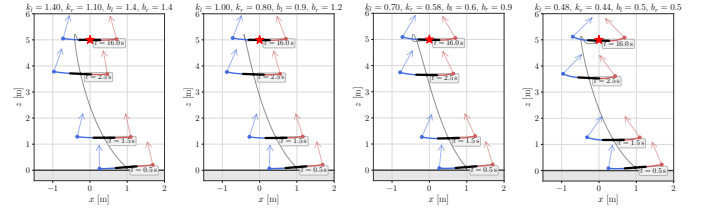
Fig. 2: State trajectories of the passive-arm Soft-PVTOL during takeoff and hover. The vehicle converges to the setpoint $(x_v^d, z_v^d) = (0, 5) \text{ m}$ and the arm curvatures q_l, q_r settle passively at a nonzero equilibrium. A lateral wind gust over $t = 6\text{--}8 \text{ s}$ is rejected, and the thrusts stay within the saturation limit T_{max} .

B. Simulation Experiments

Simulations use an RK45 integrator over $t \in [0, 16] \text{ s}$, with the prototype parameters of Table I. The initial state is $\mathbf{q}(0) = (1.0, 0.05, 0, 0.05, -0.03)^\top$, $\dot{\mathbf{q}}(0) = \mathbf{0}$, a vehicle displaced laterally with small, unequal initial curvatures. A lateral wind gust $f_{\text{wind}}(t) = 0.2 \sin(\pi(t - 6)/2) \text{ N}$ is injected on x_v for $6 < t < 8 \text{ s}$ to test robustness.

Fig. 2 shows the full state trajectories for the nominal parameter case ($k_l = k_r = 2.45 \text{ N}\cdot\text{m/rad}$, $b_l = b_r = 0.5 \text{ N}\cdot\text{m}\cdot\text{s/rad}$, Table I). The vehicle reaches the hover setpoint $(x_v^d, z_v^d) = (0, 5) \text{ m}$, the passive curvatures q_l, q_r converge to the nonzero equilibrium predicted by (5)–(6) ($q^* \approx 5.4^\circ$ for the parameters of Table I) without direct actuation, and the rotor thrusts remain smooth and within the saturation limit. The lateral gust at $t = 6\text{--}8 \text{ s}$ produces a bounded deviation in x_v and θ from which the system recovers, supporting the local controllability claim.

Fig. 3 presents takeoff snapshots for four asymmetric parameter configurations spanning a wide range of arm compliance. The vehicle reaches the hover setpoint in all of them,



(a) stiffest, well damped (b) moderate, asymmetric (c) soft, asymmetric (d) very soft, large curvature

Fig. 3: Soft-PVTOL takeoff snapshots under four passive-arm parameter configurations (k and b annotated on each panel). Despite significant stiffness and damping asymmetry, the vehicle reaches the hover setpoint $(x_v^d, z_v^d) = (0, 5) \text{ m}$ in all cases, showing robustness to arm parameter uncertainty. Snapshots at $t = 0.5, 1.5, 2.5, 16 \text{ s}$; the grey curve is the vehicle path.

including the very-soft case of Fig. 3(d), whose equilibrium curvature reaches 37° ; across the four panels q^* spans 10° to 37° . Settling quality varies, since no gains were retuned per configuration, but convergence is always achieved.

Arm compliance erodes control authority unevenly. From the prototype stiffness down to k_{min} , the arm-curvature entry of b_s, b_a loses 6% and the pitch entry 12%, whereas the thrust efficiency $\cos q^*$ falls by 35%, raising the hover thrust by 54%. The dominant cost of soft arms is therefore the lateral waste of thrust, not a weakening of the coupling to the curvature coordinates on which Theorem 1 relies. All of these depend on k alone; the damping enters only through D^* .

V. EXPERIMENTAL RESULTS

The prototype (Fig. 1) has a 3D-printed rigid body housing a Pixhawk controller and two flexible TPU arms, each carrying a brushless motor at its tip. The arms approximate the constant-curvature model, and IMUs at the tips measure pitch, pitch rate, and curvature. Tendon actuators are present but *disabled* here to emulate the passive-arm configuration. The vehicle is mounted on a test stand that constrains translation in x_v, z_v while leaving θ, q_l, q_r free. A PD controller tracks a pitch reference $\theta^d(t)$ with differential thrust $T_d = T_r - T_l$, the collective thrust held at hover, so only the antisymmetric channel Σ_a (15) is excited and q_s remains free but unexcited.

A. Pitch Tracking with Passive Arms

Fig. 4(a) shows the measured pitch θ tracking a $\pm 15^\circ$ sinusoidal reference using *only* differential thrust, with tendons disabled; a minor phase lag reflects the arm compliance, while θ stays bounded. This confirms that the thrust-to-pitch channel of Σ_a (15) is effective on hardware. The arm curvatures (Fig. 4(b)) settle at $20^\circ\text{--}24^\circ$ without tendon feedback, confirming the qualitative prediction of (5)–(6) that the passive equilibrium is nonzero. The measured deflection exceeds the value of (6) with the tabulated k , estimated from the TPU modulus rather than identified; $k \approx 0.65 \text{ N}\cdot\text{m/rad}$ reproduces the measured range, consistent with compliance concentrating

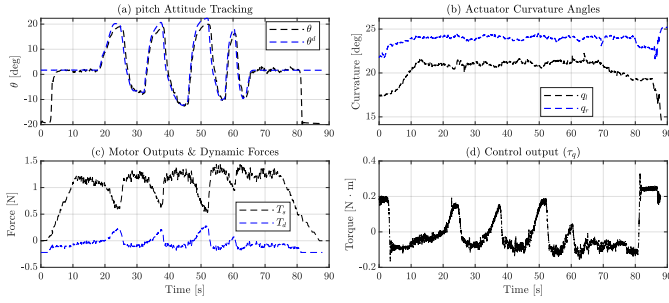


Fig. 4: Experimental validation on the test stand (translation constrained; θ, q_l, q_r free). (a) Pitch tracks θ^d via differential thrust T_d alone (no tendons). (b) Arm curvatures converge to a nonzero steady state, the $q_l - q_r$ offset reflecting the intentional stiffness asymmetry of the prototype. (c)–(d) T_d modulations and torque τ_q remain bounded.

at the layer interfaces of a printed arm. Identifying k is left to future work. Small curvature fluctuations correlated with the thrust modulations (Fig. 4(c)) directly evidence the coupling $T_i l_i (1 - \cos q_i) / q_i^2$ of (4). The $q_l - q_r$ offset reflects the intentional stiffness asymmetry of the prototype. Because the stand constrains translation, this experiment establishes accurate pitch tracking and curvature convergence with passive arms despite the asymmetry, supporting Remark 1.

B. Scope and Limitations of the Planar Analysis

The Soft-PVTOL dynamics (2) buy rigor at the price of effects they cannot represent. Out-of-plane bending tilts the thrust vectors out of the sagittal plane and generates yaw torques with no counterpart in (2); aerodynamic drag on the deformed arms and residual thrust misalignment act likewise. The supplementary video of QuadSoft [11] free flight, its three-dimensional counterpart, is consistent with this picture, including a mild yaw drift on landing. It is offered as qualitative evidence that tendon-free flight is realizable, not as validation of the planar result. Extending the controllability analysis to three dimensions, with the yaw channel included, remains open.

VI. CONCLUSIONS

We asked whether tendon actuation is necessary to fly a soft-arm aerial vehicle, i.e. whether the Soft-PVTOL remains controllable with passive, tendon-free arms. We showed that, for the considered physical system, the passive-arm Soft-PVTOL preserves local controllability at every admissible hover equilibrium outside an isolated exceptional stiffness, at which it remains stabilizable, despite dropping from four inputs to two, the enabling mechanism being the thrust-induced generalized force $T_i l_i (1 - \cos q_i) / q_i^2$ that couples the thrusts to the otherwise unactuated arm curvatures. Simulations confirmed that a simple PD controller drives the vehicle to a stable hover, and test-stand experiments confirmed accurate pitch tracking with passive arms, with measured deflections settling at a nonzero equilibrium, as the analysis predicts qualitatively. Tendon-free

operation is therefore a viable fallback rather than the intended nominal mode, at the cost of reduced control authority. Future work includes formal stability analysis, nonlinear control for low-stiffness regimes, and extension to the three-dimensional QuadSoft.

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